

Essential Summary

The Parametric Synchronization Scheme of Chaotic System

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1 Background and Motivation

Since Pecora and Carroll (1990) demonstrated synchronization of two identical chaotic systems, many schemes have been proposed: complete, phase, anti-, lag, Q - S , generalized, and projective synchronization. **However, none of these schemes constructs a parametric error vector between the drive and response systems.** This paper introduces a fundamentally different idea — **parametric synchronization**: synchronize not only the state vectors but also the *unknown response parameters* to the given drive parameters.

The practical payoff: to achieve synchronization, one need not know the response-system parameters at all, as long as the drive parameters are given.

2 Parametric Synchronization Scheme

2.1 Drive and Response Systems

Drive system with known parameters a :

$$\dot{x} = f(x, a) \cdot x + C, \quad (1)$$

Response system with unknown parameters $\hat{a}$:

$$\dot{y} = f(y, \hat{a}) \cdot y + C + u. \quad (2)$$

2.2 Parametric Error Vectors (Core Innovation)

Define two error vectors — a state error and a **parametric error**:

$$\boxed{e = y - x, \quad e_a = \hat{a} - a} \quad (3)$$

Subtracting the drive from the response yields the combined error system:

$$(\dot{e}, \dot{e}_a)^T = D(x, y, a_i) (e, e_a)^T + U, \quad (4)$$

where D is a $2n \times 2n$ matrix function, and U is a $2n$ -dimensional controller.

Definition (global parametric synchronization). Systems (1)–(2) are globally parametric synchronous if there exists U such that

$$\lim_{t \rightarrow \infty} \|e(t)\| = 0, \quad \lim_{t \rightarrow \infty} \|e_a(t)\| = 0. \quad (5)$$

That is, both states and response parameters converge to the drive states and parameters, regardless of initial conditions.

2.3 Controller Design (Nonlinear-Term Cancellation)

Partition D into linear terms and **nonlinear terms** (those depending on x_i or y_i). Construct $\tilde{D}$ from D by flipping the sign of every nonlinear term (setting linear terms to zero):

$$\tilde{d}_{ij} = -d_{ij} \quad (\text{if } d_{ij} \text{ is nonlinear}), \quad \tilde{d}_{ij} = 0 \quad (\text{otherwise}). \quad (6)$$

The controller is

$$U = \tilde{D}(e, e_a)^T + M(e, e_a)^T \quad (7)$$

with constant matrix M . Substituting into Eq. (4) yields a **linear** error system:

$$(\dot{e}, \dot{e}_a)^T = (D + \tilde{D} + M)(e, e_a)^T. \quad (8)$$

2.4 Stability Theorem

Theorem 1. If M is chosen so that $D + \tilde{D} + M$ has all eigenvalues with negative real parts, then systems (1)–(2) are globally parametric synchronous under controller (6).

Proof. By the linearization lemma, a fixed point with Jacobian having all negative-real-part eigenvalues is asymptotically stable. Since the error system is linear, the origin is a global fixed point, so $e \rightarrow 0$ and $e_a \rightarrow 0$ as $t \rightarrow \infty$.

Remark. By the Routh–Hurwitz criterion, many choices of M satisfy the theorem. If the drive parameters were unknown and the response parameters given, the procedure is analogous.

3 Application 1: Rössler System

The Rössler system with drive parameters $a = 0.2$, $b = 0.2$, $c = 5.7$:

$$\dot{x}_1 = -x_2 - x_3, \quad \dot{x}_2 = x_1 + ax_2, \quad \dot{x}_3 = b + x_3(x_1 - c). \quad (9)$$

The response system has unknown parameters $\hat{a}, \hat{b}, \hat{c}$. Following the construction, the closed-loop error system becomes linear with matrix (12) of the paper; the parameter update laws are

$$\dot{\hat{a}} = -e_1 - 2e_a, \quad \dot{\hat{b}} = -e_2 - 2e_b, \quad \dot{\hat{c}} = -e_3 - 4e_c. \quad (10)$$

4 Application 2: Hyperchaotic Rössler System

The hyperchaotic Rössler system with $a = 0.25$, $b = 3$, $c = 0.5$, $d = 0.05$:

$$\dot{x}_1 = -x_2 - x_3, \quad \dot{x}_2 = x_1 + ax_2 + x_4, \quad \dot{x}_3 = b + x_1x_2, \quad \dot{x}_4 = -cx_3 + dx_4. \quad (11)$$

The response system has four unknown parameters. The controller (17) and the 8×8 matrix M (18) from the paper render the closed-loop error system linear with all eigenvalues in the left half-plane. Figure 3 shows the hyperchaotic attractor; Figure 4 shows that states synchronize, parameters converge, and errors vanish.

5 Key Findings

1. **New synchronization paradigm:** Constructing a *parametric error vector* $e_a = \hat{a} - a$ distinguishes this scheme from all existing synchronization schemes.
2. **State and parameter synchronization simultaneously:** Both the state vectors and the unknown response parameters converge to the drive system's, as time goes to infinity.

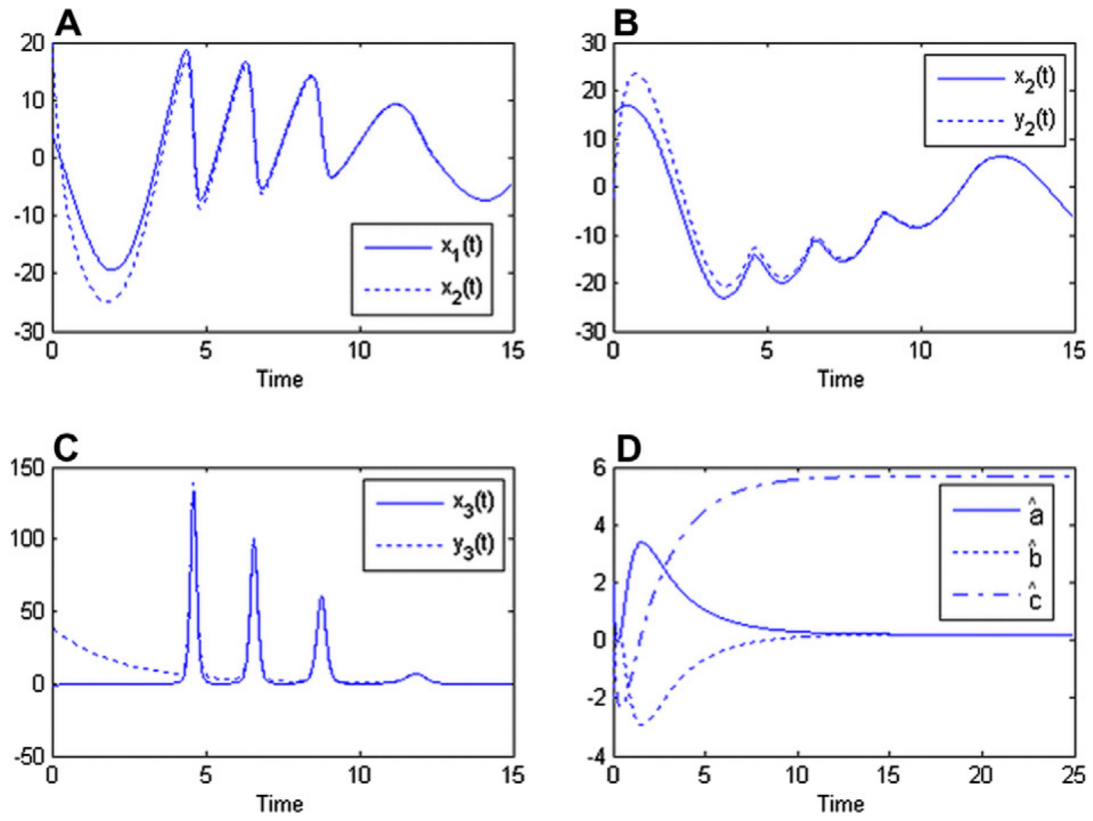


Figure 1: A–C: time responses of states for drive system and response system; D: time evolution of the unknown response parameters, converging to the drive values.

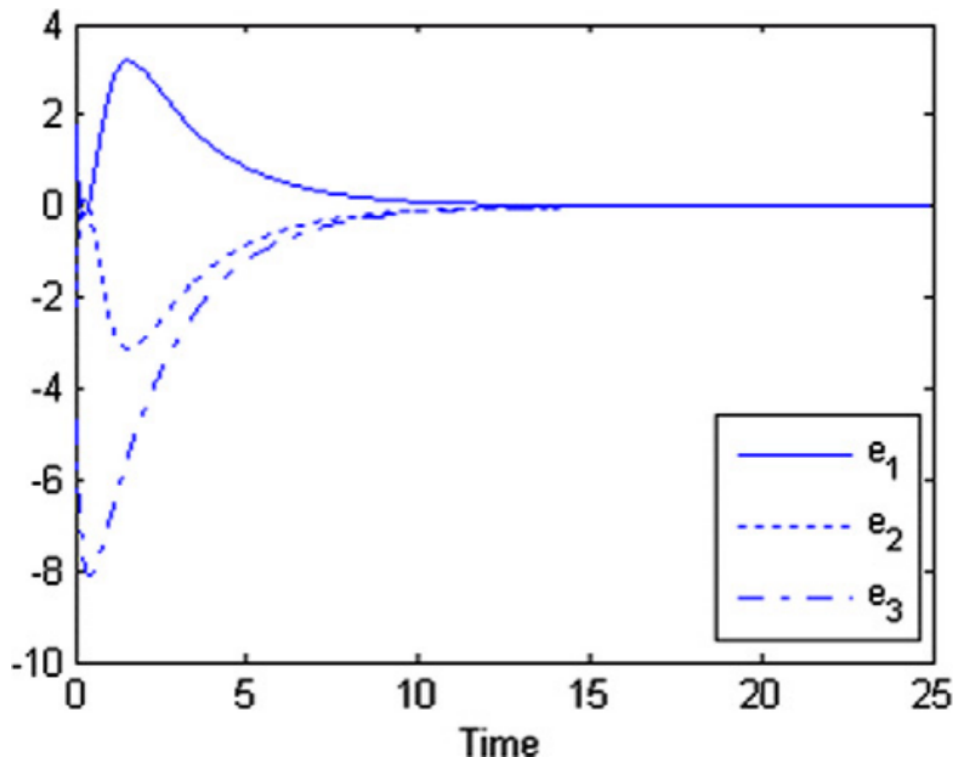


Figure 2: Time response of the synchronization errors between systems (8) and (9), converging to zero.

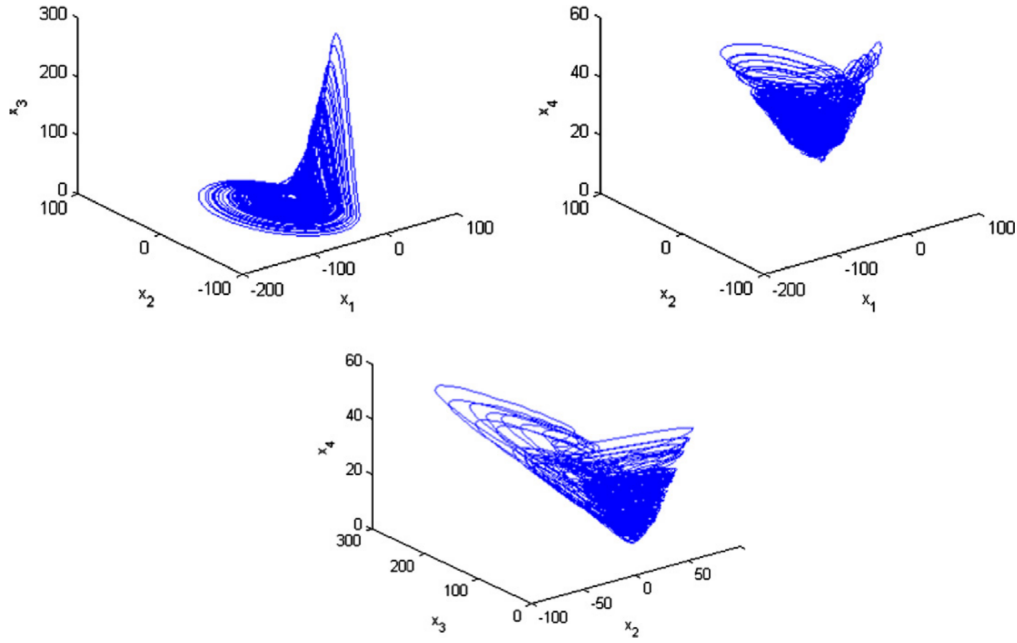


Figure 3: Three-dimensional phase portrait of the hyperchaotic Rössler system (14).

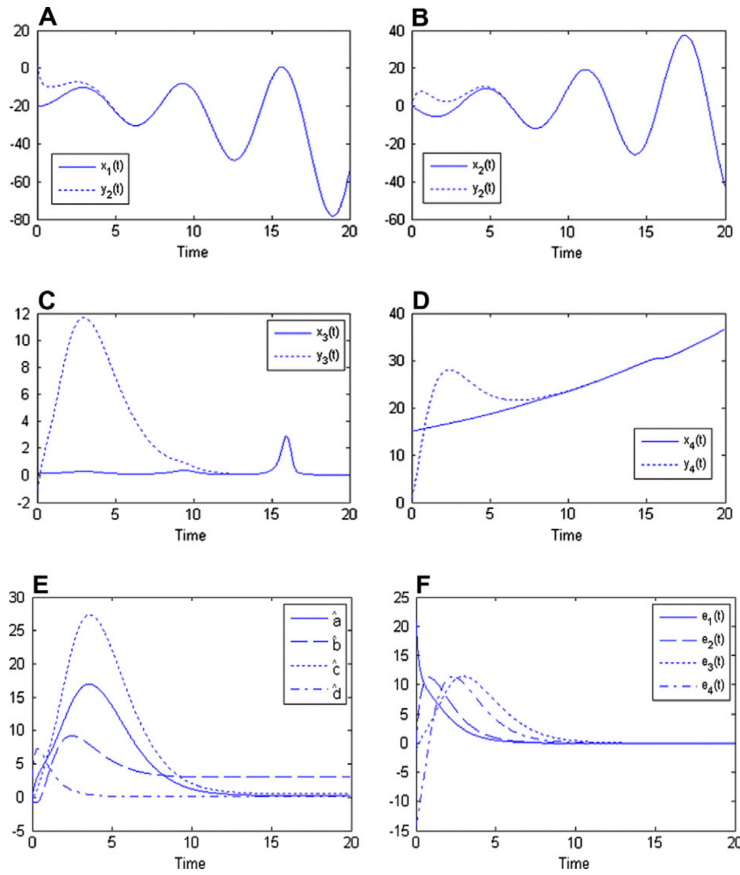


Figure 4: A–D: time responses of states; E: time evolution of the unknown response parameters; F: synchronization error states for the hyperchaotic Rössler system.

3. **No need to know response parameters:** To achieve synchronization, only the drive parameters are needed — the response parameters adapt automatically.
4. **Simple controller design:** The nonlinear-term-cancellation construction ($\tilde{D}$) plus a constant matrix M reduces the error system to a linear one, solvable via the Routh–Hurwitz criterion.
5. **Works for chaotic and hyperchaotic systems:** Validated on the Rössler system and the hyperchaotic Rössler system, with numerical simulations confirming validity.

6 Significance

This is the author’s third major work in chaos synchronization, completing a trio of 2011 papers (Acta Physica Sinica, Nonlinear Analysis RWA, and this one in Commun Nonlinear Sci Numer Simulat). Each paper develops a distinct synchronization concept — generalized projective synchronization via modified active control, generalized function projective synchronization via anti-symmetric structure, and parametric synchronization via parametric error vectors. Together they establish the “chaos synchronization” thread in the author’s research trajectory, characterized by a Lyapunov/linearization-centric controller design philosophy.

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